

UĞURCAN VURGUN

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RESEARCH INTERESTS

AI-driven healthcare interventions; natural language processing in clinical settings; machine learning for medical diagnostics; biomedical informatics; automated speech analysis for neurological conditions; large language models for healthcare applications

EDUCATION

University of Pennsylvania, Perelman School of Medicine, Philadelphia 2023 – 2027 (expected)

Master of Science (MS) in Biomedical Informatics

Institute for Biomedical Informatics

University of Pennsylvania, Philadelphia

2019 – 2024

Ph.D. in Cognitive Psychology and Psycholinguistics

Thesis Title: *From Words to Worlds: Telicity and the Malleability of Event Representations*

Interdisciplinary graduate certificate in Language and Communication Sciences (LCS)

Boğaziçi University, Istanbul

2015 – 2019

Master of Arts (MA) in Linguistics

Koç University, Istanbul

2002 – 2007

Bachelor of Arts (BA) in Business Administration

TEACHING EXPERIENCE

University of Pennsylvania College of Liberal and Professional Studies

LING 0001 Introduction to Linguistics

Summer 2023 & Summer 2024

Instructor for the course

- Integrated AI and NLP concepts, including demonstrations of large language models (LLMs)
- Developed curriculum exploring intersections of linguistics, AI, and computational methods
- Guided students in practical applications of software-oriented tools for linguistic analysis and research

University of Pennsylvania Perelman School of Medicine

BMIN 5220 Natural Language Processing For Health

Fall 2024

Teaching Assistant for the course

- Facilitated hands-on support on applying LLMs to electronic health records and clinical data analysis

Professor: Danielle L. Mowery

University of Pennsylvania School of Engineering and Applied Sciences

CIS 5300 Natural Language Processing

Fall 2024

Teaching Assistant for the course

- Guided students in implementing state-of-the-art NLP models, including transformer architectures and fine-tuning techniques

Professor: Mark Yatskar

The Wharton School

MGMT 8920 Advanced Study Program – Collaborative Innovation Program

Fall 2024

Teaching Assistant for six project teams (MBA/WEMBA students)

- Leading projects in healthcare and AI applications, including development of an AI-driven robotic dental device for Penn Dental Medicine and optimization of surgical planning using 3D-printed anatomic models for Ricoh Healthcare. Also contributing to initiatives in digital philanthropy, telecommunications data analysis, and start-up ecosystem research for various corporate partners.

Professor: Scott Snyder

The Wharton School

MGMT 8920 Advanced Study Program – Collaborative Innovation Program Spring 2024

Teaching Assistant for four project teams (MBA/WEMBA students)

- Guided projects on AI-led transformation for TATA Communications, developing partnership strategies for CHOP's Center of Advanced Diagnostics, and exploring green algorithms and AI's environmental impact for Willis Towers Watson. Also mentored a team on connected strategy for Tigo's Telecom Wholesale Business, focusing on market segmentation and process automation.

Professor: Xu (Henry) Han

The Wharton School

MGMT 8920 Advanced Study Program – Collaborative Innovation Program Fall 2023

Teaching Assistant for four project teams (MBA/WEMBA students)

- Mentored teams on AI-driven projects including intelligent software testing for Coforge and strategic AI integration for Ricoh's digital services. Guided healthcare-related initiatives to enhance plasma donor experiences at CSL Behring. Additionally, facilitated a project on AR/VR integration in Assurant's Connected Living services, focusing on improving customer experiences and market potential.

Professor: Scott Snyder

University of Pennsylvania

LING 0001 Introduction to Linguistics

Fall 2022

Teaching Assistant for two sections

Professor: Mark Liberman

LING 001 Introduction to Linguistics

Fall 2021

Teaching Assistant for two sections

Professor: Gareth Roberts

LING 001 Introduction to Linguistics

Spring 2021

Teaching Assistant for two sections

Professor: Martin Salzmann

Boğaziçi University, Istanbul

LING 101 Introduction to Linguistics

Fall 2018

Teaching Assistant

Professor: Mine Nakipoğlu

PUBLICATIONS

Donnelly, H. K., Brown, G. K., Green, K. L., Vurgun, U., Hwang, S., Schriver, E., Steinberg, M., Reilly, M. E., Mehta, H., Labouliere, C., Oquendo, M. A., Mandell, D., & Mowery, D. L. (2026). Automated safety plan scoring in outpatient mental health settings using large language models: Exploratory study. *JMIR Mental Health*, 13, e79010.

Ketchum, C. J., Vurgun, U., Wang, A., Thomas, S., Batugo, A., Falk, G. W., Lynch, K. L., Dellon, E. S., Mowery, D. L., & Lewis, J. D. (2025). Large language models combined with a single diagnostic

code detect eosinophilic esophagitis in electronic health records with high accuracy. *American Journal of Gastroenterology*, Abstract P2739.

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Aspectual processing shifts visual event apprehension. *Cognitive Science*, 48(6), e13476.

Li, C., Mowery, D. L., Ma, X., Yang, R., Vurgun, U., Hwang, S., Donnelly, H. K., Bandhey, H., Akhtar, Z., Senathirajah, Y., Sadhu, E. M., Getzen, E., Freda, P. J., Long, Q., Becich, M. J. (2024). Realizing the potential of social determinants data: A scoping review of approaches for screening, linkage, extraction, analysis, and interventions. *Journal of Clinical and Translational Science*.

Vurgun, U. (2019). The syntax of focus particles in Turkish. Unpublished master's thesis, Boğaziçi University. Advisors: Dr. H. Mine Nakipoğlu & Dr. A. Sumru Özsoy.

Vurgun, U. (2018). Book review – Contemporary Morphological Theories: A User's Guide. Thomas W. Stewart, 2015. Edinburgh: Edinburgh University Press. 192 pages. ISBN 978 07 486 9268 2. *Journal of Linguistic Research*, 29(2), 129–137.

CONFERENCE PROCEEDINGS

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Detecting event construal shifts in aspectual coercion. *Proceedings from the 46th Annual Meeting of the Cognitive Science Society*.

Vurgun, U., Heesu, Y., Landau, B., & Trueswell, J. (2024). Real-time processing of symmetrical predicates: Semantic categorization over time. *Proceedings from the 46th Annual Meeting of the Cognitive Science Society*.

Vurgun, U., Ji, Y., & Papafragou, A. (2023). Cognitive signatures of aspectual coercion. *Proceedings from the 45th Annual Meeting of the Cognitive Science Society*.

Vurgun, U., Ji, Y., & Papafragou, A. (2022). Linguistic aspect constrains event apprehension. *Proceedings from the 44th Annual Meeting of the Cognitive Science Society*.

SELECTED PRESENTATIONS

Ketchum, C. J., Vurgun, U., Wang, A., Thomas, S., Batugo, A., Falk, G. W., Lynch, K. L., Dellon, E. S., Mowery, D. L., & Lewis, J. D. (2025). Large language models combined with a single diagnostic code detect eosinophilic esophagitis in electronic health records with high accuracy. Poster presentation at the American College of Gastroenterology Annual Scientific Meeting, Phoenix, AZ.

Vurgun, U., Kaviyarasu, A., Hwang, S., Acevedo, A., Mowery, D. L., Abella, B. S., & Mitchell, O. J. L. (2024). Reliability in AI-assisted critical care: Assessing large language model robustness and instruction following for cardiac arrest identification. *AIM-FM Workshop at the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*.

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Detecting event construal shifts in aspectual coercion. Oral presentation at the 46th Annual Meeting of the Cognitive Science Society, Rotterdam, Netherlands.

Vurgun, U., Heesu, Y., Landau, B., & Trueswell, J. (2024). Real-time processing of symmetrical predicates: Semantic categorization over time. Poster presentation at the 46th Annual Meeting of the Cognitive Science Society, Rotterdam, Netherlands.

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Aspectual Coercion: A New Method to Probe Aspectual Commitments. Oral presentation at the Experiments in Linguistic Meaning (ELM) Conference, University of Pennsylvania.

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Aspectual Coercion: A New Method to Probe Aspectual Commitments. Poster presentation at the 37th Annual Conference on Human Sentence Processing, University of Michigan, Ann Arbor.

Vurgun, U., Yun, H., Landau, B., & Trueswell, J. (2024). Symmetrical or not?: A new method to probe real-time semantic categorization. Poster presentation at the 37th Annual Conference on Human Sentence Processing, University of Michigan, Ann Arbor.

Vurgun, U., Ji, Y., & Papafragou, A. (2024). Cognitive signatures of aspectual coercion. Poster presentation at the 10th Mid-Atlantic Colloquium of Studies in Meaning (MACSIM), University of Maryland, College Park.

Vurgun, U., Cho, S., Fusaroli, R., Ip, M. H. K., Pelella, M. R., Knox, A., et al. (2024). Detecting Autism through Automated Speech Analysis: The Role of Model Complexity and Dataset Balance. Poster presentation at the International Society for Autism Research Annual Meeting, Melbourne, Australia.

Vurgun, U., Cho, S., Fusaroli, R., Ip, M. H. K., Pelella, M. R., Knox, A., Schillinger, S., Hauptmann, A., Covello, M., Madya, M., Lyons, M., Lee, A., Fulop, L., Adeniyi, H., Russell, A., Hicks, G., Anello, O., Walker, K., Fiumara, J., Tunç, B., Sariyanidi, E., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Detecting Autism through Automated Speech Analysis: The Role of Model Complexity and Dataset Balance. Poster presentation at the International Society for Autism Research Annual Meeting, Melbourne, Australia.

Kaviyarasu, A., Vurgun, U., Hwang, S., Acevedo, A., Mowery, D. L., Abella, B. S., & Mitchell, O. J. L. (2024). Leveraging large language models to optimize clinical text analysis for in-hospital cardiac arrest identification. Poster presentation at the Department of Medicine Research Day, University of Pennsylvania.

Cho, S., Vurgun, U., Fusaroli, R., Ip, M. H. K., Pelella, M. R., Knox, A., Hauptmann, A., Covello, M., Madya, M., Lyons, M., Lee, A., Fulop, L., Schillinger, S., Russell, A., Hicks, G., Walker, K., Fiumara, J., Chatham, C., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Stable Acoustic Differences between Autistic and Neurotypical Children: A Longitudinal Telephony Study. Oral presentation at the International Society for Autism Research Annual Meeting, Melbourne, Australia.

Cho, S., Vurgun, U., Fusaroli, R., Ip, M. H. K., Pelella, M. R., Knox, A., Hauptmann, A., Covello, M., Madya, M., Lyons, M., Lee, A., Fulop, L., Russell, A., Hicks, G., Fiumara, J., Pandey, J., Tunç, B., Sariyanidi, E., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Automatically-derived Lexical Features Distinguish Autistic Teens from Neurotypical Peers and Teens with Other Psychiatric Conditions in a Natural Conversation Task. Oral presentation at the International Society for Autism Research Annual Meeting, Melbourne, Australia.

Ip, Martin H. K., Putnam, O., Cho, S., Vurgun, U., Lyons, M., Knox, A., Hauptmann, A., Covello, M., Madya, M., Adeniyi, H., Hicks, G., Anello, O., Walker, K., Fiumara, J., Cieri, C., Fusaroli, R., Chatham, C., Liberman, M., Schultz, R. T., Parish-Morris, J. (2024). Prosodic similarities and variation in autistic and neurotypical children: A telephone-based study of verbal counting. Oral presentation at the International Society for Autism Research Annual Meeting, Melbourne, Australia.

Vurgun, U., Cho, S., Fusaroli, R., Ip, M. H. K., Pelella, M. R., Knox, A., Hauptmann, A., Covello, M., Madya, M., Lyons, M., Lee, A., Schillinger, S., Fulop, L., Adeniyi, H., Russell, A., Fiumara, J., Hicks, G., Anello, O., Walker, K., Tunç, B., Sariyanidi, E., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Automatic classification of autism using automatic speech recognition and large language models (LLMs). Poster presentation at Meeting on Language and Autism, Durham, NC.

Cho, S., Vurgun, U., Fusaroli, R., Ip, M. H. K., Pelella, M.R., Knox, A., Hauptmann, A., Covello, M.,

Madya, M., Lyons, M., Lee, A., Schillinger, S., Fulop, L., Russell, A., Hicks, G., Fiumara, J., Pandey, J., Tunç, B., Sariyanidi, V., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Autistic teens’ word choice differs from neurotypical peers and teens with attention-deficit/hyperactivity disorder (ADHD) in a natural conversation task. Poster presentation at Meeting on Language and Autism, Durham, NC.

Ip, M. H. K., Putnam, O. C., Cho, S., Vurgun, U., Lyons, M., Knox, A., Hauptmann, A., Covello, M., Fulop, L., Russell, A., Walker, K., Miller, J., Fiumara, J., Fusaroli, R., Chatham, C., Cieri, C., Lee, A., Liberman, M., Schultz, R. T., & Parish-Morris, J. (2024). Does speech prosody differ in a socially unengaging task? Preliminary findings from autistic and non-autistic children’s verbal counting. Poster presentation at Meeting on Language in Autism, Durham, NC.

Vurgun, U., Ji, Y., & Papafragou, A. (2023). Aspectual processing and event perception. Poster presentation at the 36th Annual Conference on Human Sentence Processing, University of Pittsburgh.

Vurgun, U., Ji, Y., & Papafragou, A. (2023). Lexical aspect maps onto event apprehension. Poster presentation at the 9th Mid-Atlantic Colloquium of Studies in Meaning (MACSIM), University of Pennsylvania, Philadelphia.

Vurgun, U., Ji, Y., & Papafragou, A. (2023). Aspectual framing of events shifts online event apprehension. Poster presentation at the 98th Annual Meeting of the Linguistic Society of America, Denver, CO.

Vurgun, U., Ji, Y., & Papafragou, A. (2022). Lexical aspect maps onto event apprehension. Oral presentation at the 2nd Experiments in Linguistic Meaning Conference, University of Pennsylvania.

Vurgun, U. (2020). Both locative alternation variants in Turkish are change-of-state predicates with ResultPs. Oral presentation at the 4th Conference of Central Asian Language and Linguistics, Indiana University.

Vurgun, U. (2019). A lexical-typological approach explaining the limited existence of locative alternation in Turkish. Poster presentation at the 4th Workshop on Turkic and Languages in Contact with Turkic (Tu+4), NYU.

REVIEWING

Folia Linguistica
CogSci (The Annual Conference of Cognitive Science Society)
Journal of Clinical and Translational Science
Pacific Symposium on Biocomputing

GRANTS, FELLOWSHIPS, AND AWARDS

Benjamin Franklin Fellowship , University of Pennsylvania	2019 – 2024
University of Pennsylvania , Language and Communication Sciences Research Fund	2021, 2022, 2023, 2024
University of Pennsylvania , Integrated Language Sciences and Technology Research Fund	2022, 2023, 2024
Simons Foundation Autism Research Initiative (SFARI) Travel Award	2024
The 48th Annual Boston University Conference On Language Development , Diversity Travel Award	2023
University of Pennsylvania , GAPSA Research Travel Grant	2022
University of Pennsylvania , SAS Dean’s Travel Subvention	2022
Boğaziçi University , Research Travel Grant	2019
Vehbi Koç Fellowship , Koç University	2002 – 2007

TECHNICAL SKILLS

Programming & Analysis	R, Python, PyTorch, TensorFlow, SQL, JavaScript
Machine Learning	Scikit-learn, Keras, XGBoost
Large Language Models	HuggingFace Transformers – Fine-tuning: Transfer learning, Few-shot learning – Advanced techniques: Prompt engineering, PEFT, LoRA – Deployment: Model compression, Quantization
Software & Tools	Eye Tracking (Tobii), PCIBex, PsychoPy
Statistical Analysis	Mixed-effects modeling, Time series analysis
Linguistic Tools	TalkBank, NLTK, spaCy
Version Control	Git, GitHub
Data Visualization	ggplot2, matplotlib
Database Management	SQL, MySQL, Database Design
Web Development	JavaScript, HTML, CSS
Cloud Platforms	Microsoft Azure, Databricks

EMPLOYMENT

Amazon May 2025 – Present
Language Data Scientist

- Developing LLM-based evaluation frameworks and scalable NLP pipelines for multilingual natural language understanding across 10+ languages.
- Designing linguistically grounded quality metrics informed by psycholinguistic theory and cross-cultural language variation, advancing equitable AI evaluation for diverse populations.
- Building annotation workflows and automated scoring systems that bridge large language model capabilities with rigorous human judgment standards.

Unilever 2008 – 2018
Various Roles, Including Channel and Category Development Manager

- Progressed through multiple roles over a 10-year tenure in sales, marketing, and management, leading cross-functional teams of 200+ employees across multiple regions.
- Developed data-driven strategies for market analysis, promotional optimization, and business planning, leveraging statistical modeling to inform strategic decision-making.

PROFESSIONAL MEMBERSHIPS AND ACTIVITIES

American Psychological Association May 2024 – Present
Member

- Division 7: Developmental Psychology
- Division 33: Intellectual and Developmental Disabilities/Autism Spectrum Disorder

Wharton Founders' Club Oct 2023 – Present
Member

Family Resource Center, University of Pennsylvania Sep 2019 – Jun 2020
Student Parent Advisory Board Member